

ENTHUSIAST ADVANCE COURSE

LEADER PHASE I MEF-I (B)

TARGET : NEET-2024

Test Type : MINOR

Test Pattern : NEET (UG)

TEST DATE : 18-06-2023

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	2	3	2	1	3	4	4	2	3	4	1	4	4	4	4	1	3	4	2	1	2	3	2	4	4	2	1	4	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	1	2	1	1	1	1	2	4	1	4	1	1	4	4	3	4	1	3	4	3	4	1	2	4	4	2	2	2	4
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	4	4	3	2	2	2	2	3	1	1	1	1	2	2	4	2	1	1	2	4	4	3	2	3	3	2	3	1	4	1
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	1	3	3	1	4	4	1	3	1	4	3	4	2	2	4	4	1	4	1	4	3	2	3	3	3	3	3	2	1	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	4	3	1	2	4	2	3	1	4	1	3	2	3	3	3	3	3	4	4	1	3	1	1	3	1	4	3	3	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	3	2	1	2	3	1	3	4	4	1	3	4	2	1	2	1	3	2	2	1	4	4	1	3	1	1	2	4	3
Q.	181	182	183	184	185	186	187	188	189	190	191	192	193	194	195	196	197	198	199	200										
A.	2	4	1	4	3	4	1	1	1	4	1	4	4	3	1	4	4	1	1	2										

HINT - SHEET

SUBJECT : PHYSICS

SECTION-A

1. Ans (2)

$$\text{Least count} = \frac{\text{pitch}}{\text{No. of divisions}}$$

$$= \frac{2}{200} = 0.01 \text{ mm ;}$$

$$1 \text{ MSD} = \text{Pitch} = 2 \text{ mm}$$

$$\text{Reading} = 3 \times 2 + (46-5)(0.01) = 6.41 \text{ mm}$$

4. Ans (1)

$$\rho = \frac{m}{v}$$

% error in ρ

$$= \left(\frac{\Delta m}{m} \times 100 \right) + \left(\frac{\Delta \ell}{\ell} \times 100 \right) + 2 \left(\frac{\Delta R}{R} \times 100 \right)$$

$$= \left(\frac{0.6}{0.006} \times 100 \right) + \left(\frac{0.02}{2} \times 100 \right) + 2 \left(\frac{0.01}{1} \times 100 \right)$$

$$= 1\% + 1\% + 2\%$$

$$= 4\%$$

5. Ans (3)

$$\text{Flux through Cube} = \frac{Q}{8\epsilon_0}$$

Flux through one surface of Cube =

$$\frac{Q}{3 \times 8\epsilon_0} = \frac{Q}{24\epsilon_0}$$

6. Ans (4)

The electrostatic force due to one Cs^+ ion is balanced by diagonally opposite other Cs^+ ion. Thus, the net electrostatic force on Cl^- ion due to eight Cs^+ ions is zero.

7. Ans (4)

$$V = 0 = \frac{kx}{R} \times \frac{k(q)}{3R}$$

$$x = \frac{-q}{3}$$

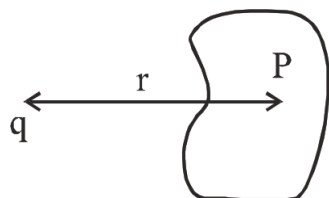
8. Ans (2)

$$\sigma \propto \frac{1}{r}$$

9. **Ans (3)**

Assertion is correct but Reason is false because under electrostatic condition free electron will be present inside the conductor.

10. **Ans (4)**



$\therefore$ Electric field inside conductor is zero

$$\therefore \vec{E}_q + \vec{E}_{\text{conductor}} = 0 \Rightarrow E_{\text{conductor}} = \frac{-Kq}{r^2}$$

11. **Ans (1)**

Potential gradient is related with electric field

according to the following relation $E = \frac{-dV}{dr}$.

$$E = -\frac{\partial V}{\partial r} = \left[-\frac{\partial V}{\partial x} \hat{i} - \frac{\partial V}{\partial y} \hat{j} - \frac{\partial V}{\partial z} \hat{k} \right]$$

$$= [\hat{i}(2xy + z^3) + \hat{j}x^2 + \hat{k}3xz^2]$$

12. **Ans (4)**

By Gauss's law $\phi = \frac{1}{\epsilon_0} (Q_{\text{enclosed}})$

$$\Rightarrow Q_{\text{enclosed}} = \phi \epsilon_0 = (-8 \times 10^3 + 4 \times 10^3) \epsilon_0$$

$$= -4 \times 10^3 \epsilon_0 \text{ Coulomb}$$

13. **Ans (4)**

Based on theory

14. **Ans (4)**

$$E_{\text{out}} = \frac{kq}{r_1^2}$$

$$E_{\text{in}} = \frac{kqr}{R^3}$$

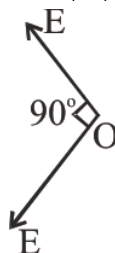
15. **Ans (4)**

$$T = 2\pi \sqrt{\frac{2m\pi\epsilon_0(2a)^3}{QQ}}$$

$$T = 2\pi \sqrt{\frac{16m\pi\epsilon_0 a^3}{Q^2}}$$

$$T = 8\pi \sqrt{\frac{m\pi\epsilon_0 a^3}{Q^2}}$$

16. **Ans (1)**



$$\therefore E_0 = \sqrt{2} E$$

$$= \sqrt{2} \cdot \frac{\sqrt{2}K\lambda}{R} \Rightarrow \frac{2K\lambda}{R}$$

22. **Ans (3)**

$$Q \text{ same } Q = C_1 V_{AD} = C_1 (V_1 - V_D)$$

$$\text{and } Q = C_2 V_{DB} = C_2 (V_D - V_2)$$

$$C_1 (V_1 - V_D) = C_2 (V_D - V_2)$$

$$C_1 V_1 + C_2 V_2 = (C_1 + C_2) V_D$$

$$V_D = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

23. **Ans (2)**

$$C = 1 \mu\text{F}$$

$$C_{\text{eq}} = C + \frac{C}{2} + \frac{C}{4} + \frac{C}{8} + \frac{C}{16} + \dots \infty$$

$$= C \left[1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots \infty \right] = C \left[\frac{1}{\left(1 - \frac{1}{2}\right)} \right]$$

$$= 2C = 2 \mu\text{F}$$

It is Geometric progression

sum of infinite terms

$$S_{\infty} = \frac{a}{1-r} \quad \text{Here } a = C, r = 1/2$$

Where $a \rightarrow$ first term, $r \rightarrow$ common ratio

24. **Ans (4)**

$$I_{\text{max}} = (\sqrt{I_1} + \sqrt{I_2})^2$$

$$= (\sqrt{I} + \sqrt{4I})^2$$

$$= 9I$$

$$I_{\text{min}} = (\sqrt{I_1} - \sqrt{I_2})^2$$

$$= (\sqrt{I} - \sqrt{4I})^2$$

$$= I$$

25. **Ans (4)**

$$\beta = \frac{\lambda D}{d}$$

$$= 6 \times 10^{-4}$$

26. **Ans (2)**

$$I = I_0 \cos^2 \theta$$

$$I = I_0 \cos^2 45^\circ$$

$$I = \frac{I_0}{2}$$

27. **Ans (1)**

Resolving limit

$$d\theta = \frac{1.22\lambda}{a} = \frac{1.22 \times 4500 \times 10^{-10}}{1} = 5.49 \times 10^{-7} \text{ rad.}$$

28. **Ans (4)**

$$x = \frac{9\lambda D}{2d} - \frac{\lambda D}{d} \quad \left\{ \begin{array}{l} D = 50 \text{ cm} \\ d = 0.15 \text{ mm} \end{array} \right.$$

$$7 \times 10^{-3} = \frac{\lambda D}{d} \left(\frac{9}{2} - 1 \right)$$

$$7 \times 10^{-3} = \frac{7}{2} \frac{\lambda D}{d}$$

$$\lambda = 600 \text{ nm}$$

29. **Ans (1)**

' I_{net} ' for incoherent source

$$I_{\text{net}} = I_1 + I_2 + I_3 + \dots$$

30. **Ans (1)**

$$\text{Angular width} = \frac{2\lambda}{a}$$

31. **Ans (4)**

Distance between the first dark fringes on either side of central maxima = width of central maxima

$$= \frac{2\lambda D}{d} = \frac{2 \times 600 \times 10^{-9} \times 2}{1 \times 10^{-3}} = 2.4 \text{ mm.}$$

32. **Ans (1)**

$$I \propto A^2$$

$$\Rightarrow \sqrt{I} \propto A$$

$$\therefore \frac{A_1}{A_2} = \sqrt{\frac{I_1}{I_2}}$$

$$= \sqrt{\frac{4}{1}}$$

$$= \frac{2}{1}$$

$$\Rightarrow A_1 : A_2 = 2 : 1$$

33. **Ans (2)**

$$\beta = \frac{\lambda D}{d}$$

$$\lambda_w < \lambda_a$$

$$\lambda \downarrow ; \beta \downarrow$$

34. **Ans (1)**

$$\beta = \frac{\lambda D}{d}$$

$$\therefore \beta \propto \lambda$$

35. **Ans (1)**

Only transverse waves undergo polarization.

SECTION-B

36. **Ans (1)**

$$n_2 = 128 \left[\frac{1000 \text{ g}}{50 \text{ g}} \right]^1 \left[\frac{100 \text{ cm}}{25 \text{ cm}} \right]^{-3}$$

$$= \frac{128 \times 10^3}{50} \times [4]^{-3}$$

$$= \frac{128 \times 10^3}{50} \times \frac{1}{64}$$

$$= 40$$

37. **Ans (1)**

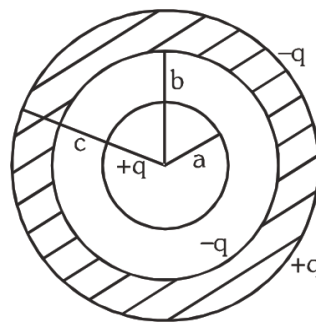
$$Q = \int_{\theta=0}^{\theta=\pi} dQ = \int_{\theta=0}^{\theta=\pi} (\lambda_0 \sin \theta) R d\theta$$

$$Q = \lambda_0 R [-\cos \theta]_0^\pi$$

$$Q = \lambda_0 R [\cos 0 - \cos \pi]$$

$$Q = 2\lambda_0 R$$

38. **Ans (2)**



(i) for $r < a$

$$E_{\text{in}} = \frac{\rho r}{3\epsilon_0} = \frac{kQr}{a^3} \quad (\text{in side field})$$

(ii) for $a < r < b$

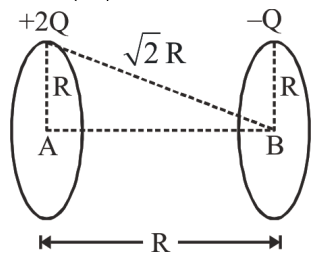
$$E = \frac{Kq}{r^2}$$

(iii) For $b < r < c$; $E = 0$ (conducting surface)

(iv) Charge on inner surface of shell = $-q$

(v) On outer surface = 0

39. Ans (4)



$$V_A = \frac{1}{4\pi\epsilon_0} \left[\frac{2Q}{R} + \frac{-Q}{\sqrt{2}R} \right] = \frac{Q(2\sqrt{2}-1)}{4\pi\epsilon_0\sqrt{2}R}$$

$$V_B = \frac{1}{4\pi\epsilon_0} \left[\frac{-Q}{R} + \frac{2Q}{\sqrt{2}R} \right] = \frac{Q}{4\pi\epsilon_0 R} [(\sqrt{2}-1)]$$

$$= \frac{Q}{4\pi\epsilon_0\sqrt{2}R} [(2-\sqrt{2})]$$

$$W_{B \rightarrow A} = q(V_A - V_B) = \frac{qQ}{4\sqrt{2}\pi\epsilon_0 R} [3\sqrt{2}-3]$$

41. Ans (4)

$$C = \frac{\epsilon_0 A}{d}; C_1 = \frac{\epsilon_0 A}{2d} = \frac{C}{2}$$

$$C_2 = \frac{5\epsilon_0 A}{2d} = \frac{5}{2}C$$

$$C_{eq} = C_1 + C_2 = 3C$$

$$\frac{\Delta C}{C} \% = \frac{2C}{C} \times 100 = 200\%$$

42. Ans (1)

$$E = \frac{V}{d}$$

$$\therefore d \uparrow E \downarrow$$

$$\therefore Q = CV \text{ and } C = \frac{\epsilon_0 A}{d}$$

$$\therefore d \uparrow C \downarrow Q \downarrow$$

43. Ans (1)

Put $k_1 = k_2 = k_3 = k_4$ and check answer

45. Ans (4)

$$V_{com} = \frac{6 \times 4 - 6 \times 3}{4 + 3} = \frac{6}{7}$$

$$C_{eq} = 7\mu F \Rightarrow U = 2.57 \mu J$$

46. Ans (3)

$$\frac{\lambda_1 D}{a} = \frac{3}{2} \frac{\lambda_2 D}{a}$$

$$\Rightarrow \lambda_2 = \frac{2}{3} \lambda_1 = \frac{2}{3} \times 660 = 440 \text{ nm}$$

47. Ans (4)

$$\left\{ \beta = \frac{\lambda D}{d} = 2 \times 10^{-4} \text{ m} \right\} \quad d' = \frac{d}{2} \quad \& \quad D' = 2D$$

$$\therefore \beta' = \frac{\lambda D'}{d'} = \frac{\lambda \times 2D}{d/2} = \frac{4\lambda D}{d}$$

$$\Rightarrow \beta' = 4\beta$$

$$\Rightarrow \beta' = 4 \times 2 \times 10^{-4} \text{ m}$$

$$\Rightarrow \beta' = 8 \times 10^{-4} \text{ m.}$$

48. Ans (1)

$$10\beta = 9.72 \text{ mm}$$

$$\Rightarrow 10 \frac{\lambda \times 100 \text{ cm}}{0.5 \times 10^{-1} \text{ cm}} = 0.972 \text{ cm}$$

$$\Rightarrow \lambda = \frac{0.972 \times 0.05}{10 \times 100} \text{ cm}$$

$$\Rightarrow 4.86 \times 10^{-5} \text{ cm}$$

49. Ans (3)

$$\theta = \frac{1.22\lambda}{a}$$

$$\Rightarrow \frac{1 \text{ mm}}{\ell} = \frac{1.22 \times 500 \text{ nm}}{3 \text{ mm}}$$

$$\Rightarrow \ell = \frac{3 \times 10^{-3} \text{ m} \times 10^{-3} \text{ m}}{1.22 \times 500 \times 10^{-9} \text{ m}} = \frac{3}{0.61} \text{ m}$$

$$\Rightarrow \ell \simeq 5 \text{ m}$$

50. Ans (4)

$$\text{Let intensity at 1 slit} = x. \therefore I_0 = 4x$$

$$\text{Now, when } \Delta x = \frac{\lambda}{6}$$

$$\Delta \phi = \frac{2\pi}{\lambda} \times \frac{\lambda}{6} = \frac{\pi}{3}$$

$$\therefore \cos \Delta \phi = \frac{1}{2}$$

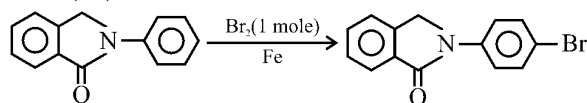
$$\therefore I = x + x + 2x \times \frac{1}{2} = 3x \quad \therefore I = 3x$$

$$\therefore \frac{I}{I_0} = \frac{3x}{4x} = \frac{3}{4}$$

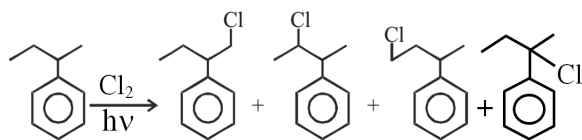
SUBJECT : CHEMISTRY

SECTION-A

55. **Ans (4)**



58. **Ans (2)**



65. **Ans (2)**

$$z = 115 \text{ [Group 15]}$$

$$\text{GEC} = ns^2 np^3$$

78. **Ans (1)**

Milk is emulsion of liquid fat (oil) in liquid medium (water)

79. **Ans (2)**

CHCl_3 & CH_3COCH_3 show negative deviation due to H-Bonding.

80. **Ans (4)**

Lower the value of K_H . Higher will the solubility of gas.

81. **Ans (4)**

$$\text{Osmotic pressure } (\pi) = iCRT$$

$$\pi \propto i \times C$$

82. **Ans (3)**

$$Y_A = \frac{P_A^0 \times A}{P_B^0 X_B + P_A^0 X_A} = \frac{100 \times 0.4}{100 \times 0.4 + 200 \times 0.6}$$

$$= \frac{40}{40 + 120} = \frac{40}{160} = \frac{1}{4} = 0.25$$

83. **Ans (2)**

$$i = 1 + (n-1) \propto n = 4$$

$$i = 1 + (4-1) \times 0.9$$

$$i = 1 + 2.7 = 3.7$$

84. **Ans (3)**

$$\text{Molarity of pure water} = \frac{1000}{18} = 55.5$$

85. **Ans (3)**

$$N = \text{molarity} \times v.f$$

$$v.f(\text{HCl}) = 1$$

$$v.f(\text{H}_2\text{SO}_4) = 2$$

$$v.f(\text{H}_3\text{PO}_4) = 3$$

SECTION-B

99. **Ans (1)**

$$\text{SCC} = 1$$

$$\text{FCC} = 4$$

$$\text{BCC} = 2$$

→ No of atoms in 1 unit cell.

SUBJECT : BIOLOGY-I

SECTION-A

107. **Ans (1)**

NCERT-XII, Pg. # 245

131. **Ans (1)**

NCERT XII Page No. # 103

134. **Ans (3)**

NCERT-XII, Pg. # 108

SECTION-B

136. **Ans (3)**

NCERT Pg. # 253, 254, 255

137. **Ans (3)**

NCERT Pg. # 239 (H)

149. **Ans (3)**

NCERT XII (E) Pg. 114 Diagram 6.12

SUBJECT : BIOLOGY-II

SECTION-A

154. **Ans (1)**

NCERT 12th Pg. No. 131

160. **Ans (4)**

NCERT-XII, Pg. # 140

161. **Ans (1)**

NCERT XII Pg # 126

163. **Ans (4)**
NCERT XII Pg # 128
164. **Ans (2)**
NCERT XII Pg # 129
165. **Ans (1)**
NCERT XII Pg # 129
166. **Ans (2)**
NCERT XII Pg # 137
167. **Ans (1)**
NCERT XII Pg # 137
169. **Ans (2)**
NCERT XII Pg # 139
170. **Ans (2)**
NCERT XII Pg # 142
171. **Ans (1)**
NCERT XII Pg # 134

172. **Ans (4)**
NCERT XII Pg # 138

SECTION-B

187. **Ans (1)**
NCERT (XII) Pg#127/137(H) para 7.1
189. **Ans (1)**
Module
192. **Ans (4)**
NCERT Pg#132
193. **Ans (4)**
NCERT XII Pg # 130
195. **Ans (1)**
NCERT Pg # 131, 132